HW9 Surface Integrals

(1a)

$x = u\cos v$ $\quad 0 \le u \le 2$

$y = u\sin v$ $\quad 0 \le v \le 2\pi$

$z = 4 - (u^2\cos^2 v + u^2\sin^2 v) = 4 - u^2$

$\Rightarrow \vec{r}(u,v) = [u\cos v, u\sin v, 4-u^2]$

$\vec{r}_u = [\cos v, \sin v, -2u]$

$\vec{r}_v = [-u\sin v, u\cos v, 0]$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = [2u^2\cos v,\ 2u^2\sin v,\ u\cos^2 v + u\sin^2 v]$$
$$= [2u^2\cos v,\ 2u^2\sin v,\ u]$$

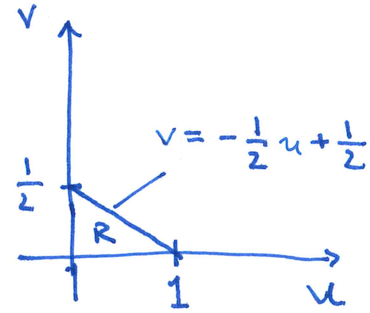
$$\vec{F}(\vec{r}(u,v)) = [u\cos v, u\sin v, 4-u^2]$$

$$\text{Flux} = \iint_S (2u^3\cos^2 v + 2u^3\sin^2 v + 4u - u^3)\,dA$$
$$= \int_0^{2\pi}\int_0^2 (2u^3 + 4u - u^3)\,du\,dv$$
$$= \int_0^{2\pi}\int_0^2 (u^3 + 4u)\,du\,dv$$
$$= 2\pi\left[\frac{u^4}{4} + 2u^2\right]_0^2$$
$$= 2\pi(4+8)$$
$$= \boxed{24\pi}$$

(1b)

$\vec{N} = [4,8,1]$ $\qquad$ $\vec{r}(u,v) = [u, v, 4-4u-8v]$

$\vec{F}(\vec{r}(u,v)) = [u, v, 4-4u-8v]$



$$\text{Flux} = \int_0^1 \int_0^{-\frac{1}{2}x+\frac{1}{2}} [\cancel{4u} + \cancel{8v} + 4 - \cancel{4u} - \cancel{8v}]\, dv\,du$$

$$= \int_0^1 \int_0^{-\frac{1}{2}x+\frac{1}{2}} 4 \, dv\,du$$

$$= 4\left(\tfrac{1}{2}\right)^2 = \boxed{1}$$

(1c)

$x = 2\sin u \cos v$ $\qquad$ $0 \le u \le \pi$

$y = 2\sin u \sin v$ $\qquad$ $0 \le v \le 2\pi$

$z = 2\cos u$

$\vec{r}(u,v) = [2\sin u \cos v, 2\sin u \sin v, 2\cos u]$

$\vec{r}_u = [2\cos u \cos v, 2\cos u \sin v, -2\sin u]$

$\vec{r}_v = [-2\sin u \sin v, 2\sin u \cos v, 0]$

$$\vec{N} = [4\sin^2 u \cos v, +4\sin^2 u \sin v, 4\cos u \sin u \cos^2 v + 4\cos u \sin u \sin^2 v]$$

$$= [4\sin^2 u \cos v, +4\sin^2 u \sin v, 4\cos u \sin u]$$

$\vec{F}(\vec{r}) = [2\sin u \cos v, 2\sin u \sin v, 2\cos u]$

$$\text{Flux} = \int_0^{2\pi} \int_0^{\pi} 8\sin^3 u \cos^2 v + 8\sin^3 u \sin^2 v + 8\cos^2 u \sin u \, du\,dv$$

$$= \int_0^{2\pi} \int_0^{\pi} [8\sin^3 u + 8\cos^2 u \sin u]\, du\,dv$$

$$= \int_0^{2\pi} \int_0^{\pi} 8\sin u\,[\sin^2 u + \cos^2 u]\, du\,dv$$

$$= 8\int_0^{2\pi} \int_0^{\pi} \sin u \, du\,dv$$

$$= 8(2\pi)(-\cos u)\Big|_0^{\pi} = 16\pi(1+1) = \boxed{32\pi}$$